J. Dairy Sci. 109:7100–7113
https://doi.org/10.3168/jds.2025-27223

Relationship between dry matter intake, milk performance, and energy efficiency from dairy cows offered grass-only, grass–white clover, and total mixed ration feeding systems

E. Fitzpatrick,[1,2,3]* L. Delaby,[4] S. McAuliffe,[1] M. A. Hurley,[1,5] N. Galvin,[1] T. J. Gilliland,[2] and D. Hennessy[1,6]

[1]Teagasc, Animal & Grassland Research and Innovation Centre, Moorepark, Fermoy, Co. Cork, P61 C996, Ireland
[2]Institute for Global Food Security, Queen's University Belfast, BT7 1NN, Northern Ireland
[3]Teagasc, Environmental Research Centre, Johnstown Castle, Co. Wexford, Y35 TC97, Ireland
[4]INRAE, Institut Agro, UMR 1348, Physiologie, Environnement et Génétique pour l'Animal et les Systèmes d'Elevage, 35590 Saint Gilles, France
[5]Munster Technological University Kerry, Tralee, Co. Kerry, V92 CX88, Ireland
[6]School of Biological, Earth and Environmental Sciences, University College Cork, Distillery Fields, North Mall, Cork, T23 N73K, Ireland

ABSTRACT

Dry matter intake is one of the most limiting factors for milk production in pasture-based dairy production systems. The inclusion of white clover (*Trifolium repens* L.) in grass swards can increase voluntary DMI through improved sward nutritive quality and lower resistance to breakdown in the rumen. Feeding a TMR can support greater milk production from increased daily DMI, and greater control of feed intake quality. The objective of this study was to determine the relationship between DMI, milk production, and energy efficiencies for dairy cows consuming different diets. The 3 treatments imposed were perennial ryegrass (PRG; *Lolium perenne* L.) receiving 250 kg N/ha (GR), PRG and white clover receiving 150 kg N/ha (CL), and TMR. Dry matter intake was estimated 17 times over the duration of the study (2015–2021). Simultaneously, milk production, production efficiencies, energetic efficiencies, and feed quality were also measured. Significant increases in milk yield production (+5.0 kg), and DMI (+2.8 kg) were observed when cows consumed the TMR diet compared with the pasture-based diets. In terms of energetic efficiencies, greater energy (+2.84 unité fourragère lait [UFL]) was available for milk production after maintenance for the TMR treatment compared with GR and CL, whereas TMR and CL cows required similar energy to produce 1 kg standard milk yield. All treatments had similar energy (148.64 g milk solids/UFL intake) available for milk solids production after accounting for maintenance. Cows fed the TMR diet consumed an additional 2.30 kg DMI daily compared with the CL cows. The cows grazing the CL swards consumed, on average, 1.03 kg greater total DMI compared with the GR cows. This translated into greater daily milk (+1.2 kg) and milk solids (0.12 kg) yield compared with the GR treatment. At a white clover inclusion rate of 31.3% and 37.7% in July and September, respectively, the herbage NDF was significantly reduced compared with GR. The current study highlights the benefits of a TMR and PRG-white clover feeding system for increasing DMI, milk production, and energy efficiencies compared with PRG only.
Key words: dry matter intake, white clover, milk production, TMR, milk production efficiency

INTRODUCTION

In dairy cow feeding systems, one of the main challenges is providing an energy-dense diet that does not compromise the rumen ecosystem, animal welfare, or milk production performance (Krause and Oetzel, 2006; Zebeli et al., 2008). Different milk production systems, such as indoor feeding and pasture-based systems, exist in different regions, with diverse advantages and consequences in terms of inputs.

One of the main limiting factors for milk production in pasture-based systems is low DMI (Bargo et al., 2003). In contrast, confinement systems feed a well-balanced TMR diet, which has the potential to maximize milk production, with the option to adjust the diet to meet the cows nutritional requirements depending on age, stage of lactation, and so on (Kolver and Muller, 1998; Bargo et al., 2002). Feeding dairy cows a TMR diet composed of a mix of grass and maize silage and concentrates supports higher milk production per cow through increased daily DMI and greater control of feed intake quality (Charlton et al., 2011). The feed efficiency of pasture-fed cows can be compromised due to limited energy intake (Kolver

Received July 9, 2025.
Accepted February 19, 2026.
*Corresponding author: ellen.fitzpatrick@teagasc.ie

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

and Muller, 1998). Under grazing conditions, dairy cows generally have lower DMI and milk yield than cows in confinement systems; however, they typically incur lower production costs per kilogram of milk and milk components, and exert less environmental pressure (Dillon et al., 1995; Peyraud et al., 2010; Macdonald et al., 2011; Lorenz et al., 2019).

White clover (*Trifolium repens* L.) is an important sward species in temperate regions because it allows for the reduction of chemical nitrogen (N) fertilizer application and increases the nutritional value of the sward (Egan et al., 2018; McClearn et al., 2019b). Dry matter intake can be influenced by sward composition, and specifically by white clover content, as white clover is more palatable and digestible than perennial ryegrass (**PRG**; *Lolium perenne* L.; Ribeiro Filho et al., 2003; Egan et al., 2018; McClearn et al., 2019a). Rutter (2006) reported that when grazing cows are offered the choice of PRG or white clover swards, they have a 70% partial preference for white clover. This preference can be attributed to a faster rate of rumen passage, a lower resistance to breakdown during the eating and rumination processes and the lower fiber fraction content associated with white clover compared with PRG (Minson 1990; Enriquez-Hidalgo et al., 2014a; McClearn et al., 2019b). As a result, many studies report that including white clover in swards of PRG can increase DMI (Ribeiro Filho et al., 2003; Egan et al., 2018; McClearn et al., 2021); others, however, have reported no effect of white clover inclusion on DMI (Ribeiro Filho et al., 2005; Enriquez-Hidalgo et al., 2014a). Nonetheless, other factors that may influence DMI for cows consuming herbage with or without white clover include sward clover content, pregrazing herbage mass and postgrazing sward height (Ribeiro Filho et al., 2003; Ribeiro Filho et al., 2005; Enriquez-Hidalgo et al., 2014a).

Differences in energy maintenance requirements related to walking and grazing activity (Bargo et al., 2002) and intake of nutrients (Kolver and Muller, 1998) are the primary factors limiting DMI and constraining milk production of cows consuming pasture versus cows consuming a TMR. Previous research has indicated that the lower DMI of pasture-fed cows, compared with TMR-fed cows, is due to 3 main factors influencing ingestive behavior: DM per bite, bite rate, and grazing time (Kolver and Muller, 1998). Including white clover in the herbage has been proven to increase milk production and efficiency compared with a grass-only diet (McClearn et al., 2021). McClearn et al. (2019b) concluded that cows consuming white clover swards supported an additional 156 kg of milk solids (**MSol**)/ha per year compared with PRG-only swards. Meanwhile, O'Neill et al. (2011) reported that TMR-fed cows produced a daily milk yield of 26.96 kg/cow, whereas cows consuming PRG-only diet produced only 18.98 kg/cow during 2 measurement periods in early lactation.

Although feeding a TMR diet ensures animals achieve their maximum genetic milk production, the implementation of TMR systems in temperate regions such as Ireland may not always make economic sense, because it does not exploit the large quantities of cheap feedstuff available in the form of grazed grass (O'Donovan et al., 2011). Previous studies have compared grass-only to grass-clover diets (McClearn et al., 2021; Murray et al., 2024) or grass-only to TMR diets (Kolver and Muller, 1998). However, very few studies have been carried out simultaneously comparing DMI and efficiencies from grass-only, grass–white clover, and TMR systems on a similar type of cow. Thus, the objective of this experiment was to determine the consequences of these dairy feeding systems on milk production, DMI, and energy efficiency when a similar type of cow was used.

MATERIALS AND METHODS

A database was assembled from 6 yr of a systems study conducted at Teagasc, Moorepark, Fermoy, Co. Cork, Ireland, from 2015 to 2021. Data from 2018 were excluded due to a drought in summer 2018, which resulted in high quantities of concentrate and grass silage being fed to the pasture-based treatments. Fifty-four spring calving dairy cows were selected each year from the Moorepark herd and allocated to 1 of 3 treatments: PRG-only (**GR**), PRG and white clover (**CL**), or TMR. A mixture of Holstein-Friesian, Friesian, and Friesian × Jersey, and primiparous and multiparous cows were assigned to 1 of the 3 treatment groups based on calving date (February 16 ± 5.3 d), lactation number (2.8 ± 0.26), 2-wk pre-experimental milk yield (24.2 ± 1.99 kg), and 2-wk pre-experimental MSol yield (2.17 ± 0.195 kg). Cows remained in their treatment groups for the entire lactation each year. Cows were randomized to a new treatment each year so no cumulative treatment effect was recorded. All experimental procedures on the animals in this study were approved by the Health Products and Regulatory Authority of Ireland under the project authorizations AE19132/P019, AE19132/P095 and AE19132/P138.

Feeding System and Cow Management

The TMR group were housed in cubicle accommodation year-round. The TMR was composed of grass silage, maize silage, and concentrate. The composition of the TMR varied between years depending on the maize silage and grass silage nutritive values. New feed was mixed daily and allocated every morning using a Keenan diet feeder (Keenan Ltd., Co. Carlow, Ireland). The TMR was fed via a Griffith Elder feeding system (Griffith Elder

Ltd., Bury St. Edmunds, Suffolk, UK) in 2015 and 2016 and via roughage intake control system feed bins (Hokofarm Group B.V.) from 2017 to 2021; both recorded daily individual cow intake. The target daily DMI of the TMR was 22.9 kg/d, on average, over the 7 yr of the experiment; this was based on the predicted intake of the cows plus 10% for refusals.

For the 2 grazing groups, a farm systems experiment was conducted over 6 grazing seasons from February 2015. The treatments were cows grazing outdoors on grass-only swards receiving 250 kg N/ha (GR) or grass-clover swards receiving 150 kg N/ha (CL). The GR swards were a 50:50 PGR mixture of AstonEnergy (tetraploid) and Tyrella (diploid) sown at a rate of 27.2 kg/ha. The CL sward contained the same PGR mixture as the GR, plus a 50:50 blend of Chieftan and Crusader medium leaf white clovers sown at a rate of 5 kg/ha. Swards were sown in July 2012, June 2013, June 2016, May 2019, July 2019, and July 2021. Following reseeding, swards were sprayed with a postemergence herbicide to control weeds. A clover-safe postemergence herbicide was used for the CL swards. The grazing treatment farmlets were stocked at 2.74 cows/ha. The GR and CL treatments closed approximately 25% of the farmlet area for first cut silage in late May and 20% for second cut in July. Each farmlet was managed independently to optimize herbage performance. Weekly farm walks were conducted to monitor average farm cover using PastureBase Ireland (**PBI**; Hanrahan et al., 2017), the primary decision-support tool for grazing management. When PBI identified herbage surpluses, excess forage was removed as baled silage. If an individual grazing treatment had a feed deficit, cows were supplemented with silage produced within that treatment. If both GR and CL treatments had a deficit, all cows received concentrate supplementation.

Nitrogen fertilizer was applied as urea (46% N) until the end of March and as calcium ammonium nitrate (27% N) from April to mid-September after each grazing from 2015 to 2019, and as protected urea (46% N) in 2020 and 2021. Nitrogen was applied in accordance with the Nitrates Directive (91/676/EEC; European Commission, 1991).

Cows were turned out to graze day and night from early February onward as weather conditions allowed. Swards were rotationally grazed from February to November to achieve an average of 9 rotations per year. The target pregrazing herbage mass (4 cm above ground level) mid-season (April to mid-August) was 1,300 to 1,500 kg of DM/ha. During the DMI estimation periods, a daily herbage allowance of 18 kg of DM/cow per day was offered to the cows in 2015, 2016, and 2017; 17 kg of DM/cow per day was offered, with a 0.89 kg of DM/cow per day individual concentrate allowance in 2019, 2020, and 2021. The grazing cows were fed an average of 501 kg of concentrate annually during lactation.

Sward Measurements and WC Contribution

Sward data were collected on all paddocks before grazing throughout the DMI estimation periods. Pregrazing herbage mass, pre- and postgrazing sward height, and sward density were measured as described by Egan et al. (2018). The grass supply for each treatment was independently managed by weekly monitoring of average farm cover (O'Donovan et al., 2002). The online application PBI (Hanrahan et al., 2017) was used to aid decision-making.

Before every grazing, a composite herbage sample was removed using a Gardena hand shears (Accu 60; Gardena International GmbH, Ulm, Germany). Samples were taken from at least 15 random locations within the paddock and cut to approximately 4 cm to reflect the target postgrazing sward height. This was then mixed and two 70-g subsamples removed and separated by hand into grass (PRG and weed grasses) and white clover fractions and dried at 90°C for 15 h to determine the proportions on a DM basis.

Chemical Analysis

The individual components of the TMR diet were sampled daily during the DMI estimation periods. The DM of the maize silage, grass silage, and concentrate was determined by drying the ingredients individually at 90°C for 15 h. The TMR formulation was modified weekly based on DM content to ensure consistent composition throughout lactation. Samples of each of the TMR ingredients were analyzed and prepared similarly to the herbage samples and analyzed using wet chemistry as described below.

Herbage representative of that selected by the cows was collected daily during the DMI measurement period. Each sample was manually collected with a Gardena hand shears as described by Ganche et al. (2013), and the sample was harvested taking the previous day's postgrazing sward height into account. These samples were stored at −18°C, bowl-chopped (Muller, Type MKT 204 Special, Saabrücken, Germany), freeze-dried at −50°C, and milled through a 1-mm screen using a Cyclotech 1093 Sample Mill (Foss, DK-3400 Hillerød, Denmark) and stored before analysis by wet chemistry. Samples were analyzed for DM, OM digestibility (**OMD**) with the in vitro neutral detergent cellulose method (Morgan et al., 1989). Neutral detergent fiber and ADF concentrations were measured using a fiber analyzer (Ankom Technology, Macedon, NY) following the method of Van Soest et al. (1991). Amylase and sodium sulfite were included during the NDF determination. Ash concentration was determined by incinerating a subsample in a muffle furnace at 500°C for 12 h, and CP concentration was quantified using a N analyzer (FP-428; Leco Australia Pty Ltd., Castle Hill, New South Wales, Australia) following AOAC method 990.03 (AOAC, 1990).

Animal Measurements

Milking took place twice daily at 0700 and 1530 h. The TMR cows were the first cows to enter the milking parlor, followed by the CL and GR cows, at each milking event. Individual cow milk yield was recorded automatically at each milking (Dairymaster, Causeway, Co. Kerry, Ireland). Milk composition (fat, protein, and lactose) was measured weekly from a successive evening and morning milk sample, using infrared spectrophotometry (Milkoscan 203 [DK-3400], Foss Electric, Hillerød, Denmark). Milk solid production was the total of the fat and protein yields from these measurements. Fat- and protein-corrected milk (**FPCM**) yield (Sjaunja, 1990) was calculated as follows: [(0.337 + 0.116 × milk fat % + 0.06 × milk protein %) × milk yield kg].

Body weight was determined by weighing cows weekly using an electronic portable scale and Winweigh software package (TRU-test Limited, Auckland, New Zealand). Body condition score was estimated by an experienced independent observer on a scale of 1 (emaciated) to 5 (extremely fat), with increments of 0.25 (Edmundson et al., 1989). The reproductive management was similar for all cows irrespective of feeding treatment.

Dry Matter Intake

Individual DMI was estimated for GR and CL cows in mid-May, mid-July, and mid-September using the *n*-alkane technique (Mayes et al., 1986, as modified by Dillon, 1993). Egan et al. (2018) and McClearn et al. (2021) used this method for the estimation of DMI for cows grazing grass-only and grass-clover swards. All cows were dosed twice daily, before morning and evening milking, with a paper bullet (Carl Roth, GmbH, Karlesruhe, Germany) containing 500 mg of dotoriacontane (C_{32}-alkane) for 12 d consecutively. From d 7 to d 12 of the dosing period, a fecal sample was collected from each cow before both morning and evening milking and stored at −18°C. Samples were later thawed and buked (12 g of each collected sample) on an individual cow basis, dried at 60°C and milled through a 1-mm screen. The diet of the animals was also sampled during the period of feces collection. During the DMI measurement periods, 2 herbage samples, representative of that grazed (accounting for the previous day's postgrazing sward height) were collected manually, daily, using a Gardena hand shears (as described previously) and stored at −18°C. These were then bowl-chopped, freeze-dried, and milled through a 1-mm screen, before chemical analysis for alkane content. Cows were fed 0.89 kg DM of concentrated per cow during the DMI estimation periods in 2019, 2020, and 2021. No silage was fed during the DMI measurement periods, making grazed pasture the main component of the diet. The ratio of grass C33 (tritriacontane) to dosed C32 was used to estimate DMI using the following equation (Mayes et al., 1986):

$$\text{DMI (kg of DM/d)} = \frac{\left(F_i/F_j\right)\times D_j}{H_i - \left[\left(F_i/F_j\right)\times H_j\right]},$$

where H is the alkane concentration of the herbage (mg/kg DM), F is the alkane concentration of feces (mg/kg DM), i is the natural or internal marker alkane with odd number carbon atoms (C_{33}), j is the synthetic external marker alkane with even number of carbon atoms (C_{32}), and Dj is the daily dose C_{32} (mg/d) alkane.

Dry matter intake was also recorded for the TMR group during the same periods. The electronic Griffith Elder or roughage intake control feeding system recorded the fresh weight of TMR consumed by each cow daily and by analyzing the DM of a subsample, the daily DMI of each cow was calculated. The different methods of DMI estimation applied in the current study were

Database Development and Production Efficiencies

Data from 277 dairy cows were available for analysis, of which 73 were primiparous and 204 were multiparous. For each week of DMI measurements, the net energy value for maintenance, growth, and milk production were calculated based on the INRA net energy system (Faverdin et al., 2011), where 1 unité fourragère lait (**UFL**) of energy is the net energy content of 1 kg barley, equivalent to 1,700 kcal. Energy balance was defined as the difference between energy intake and energy required for milk production, maintenance, and BW change (Faverdin et al., 2007). Measures of production efficiencies included energy (UFL) intake, energy balance, total DMI (**TDMI**)/100 kg BW, standard (4.0% fat and 3.1% protein content) milk yield (**STDMY**)/100 kg BW, MSol/100 kg BW, and MSol/kg TDMI. The energetic efficiencies (as per INRA, 2010) examined included UFL available for milk production after accounting for maintenance and gestation, UFL required per kilogram of STDMY and per kilogram of MSol, and UFL intake per kilogram of MSol after accounting for maintenance (similar to Evers et al., 2023).

Obvious data errors or missing data of the population for milk production and DMI traits were discarded, similarly to (Evers et al., 2023; Coffey et al., 2017). Any animals that had an energy balance in the threshold of −5 to +6 UFL/d during a measurement period were suitable and included in the analysis. Animals outside of this range were excluded because the DMI was deemed to be too high or low to be included in the analysis. Following edits and exclusions, the final dataset contained records from 231 individual animals for run 1 out of a total of 270, 277 individual animals for run 2 out of a total of 324, and 247 individual animals for run 3 out of a total of 324.

There is no DMI data included from 2018 due to poor (cold or wet) weather conditions in the spring, and a prolonged period of drought in June and July in which grass growth was curtailed and was therefore not available in adequate supplies to measure DMI without silage supplementation. Meteorological data collected at the experimental site in 2018 revealed that air temperatures were 1.9°C and 2°C lower in February and March, respectively, and 2.0°C and 1.3°C greater in June and July, respectively, compared with the 10-yr average. Rainfall was also affected during the same period, in which average rainfall was 47 mm lower in June and 20 mm lower in July compared with the 10-yr average, there was also 102 mm greater rainfall in April in 2018. No DMI measurements took place in May 2020 due to COVID-19 restrictions.

Statistical Analyses

Average daily milk yield; MSol yield; fat, protein, and lactose content; pasture DMI; TDMI; BW; BCS; FPCM energy intake; energy balance; and the milking production efficiencies were all analyzed using PROC MIXED in SAS 9.4 software (SAS Institute Inc., Cary, NC). The effects of treatment (TMR, GR, and CL), year (2015, 2016, 2017, 2019, 2020, and 2020), parity (1, 2, ≥3), DMI measurement period (May, June, and September) and the associated interactions (DMI measurement period × treatment) were included in the model as fixed effects. Dry matter intake measurement period was the repeated measure, cow was considered the experimental unit and cow-year was the random effect in the model. Tukey's test was used to measure differences between means. Significance was determined at $P < 0.05$, and a tendency at $P < 0.10$.

Grazing parameters and sward and TMR nutritive value characteristics were analyzed using PROC MIXED in SAS 9.4 software (SAS Institute Inc., Cary, NC), with the effects of year, treatment, DMI measurement period and associated interactions included as fixed effects in the model. Dry matter intake measurement period was used as the repeated measure. Data are presented as LSM with the SE. Tukey's test was used to measure differences between means. Significance was determined at $P < 0.05$, and a tendency at $P < 0.10$.

RESULTS

Sward Measurements and Feed Analysis

Table 1 shows the pregrazing herbage mass, pre- and postgrazing sward height, sward white clover content, and chemical composition of the herbage offered during the DMI measurement periods. In May and September, postgrazing sward height was higher ($P < 0.001$) for GR swards compared with CL (4.22 vs. 4.08 cm and 4.16 vs. 3.97 cm, respectively), whereas no difference was observed in July. The lowest postgrazing sward height for the CL treatment was observed in September (3.97 cm), this coincided with the highest sward white clover content (37.7%). This was the peak of a significant ($P < 0.001$) and progressive rise in clover content from May (18.9%), through July (31.3%), with an overall average of 29.3% across the 3 periods.

Herbage OMD did not differ significantly between treatments in May and September. In September, OMD was significantly greater ($P < 0.001$) for the GR herbage compared with the CL herbage. There were no differences in sward CP content noted between treatments in May or September. However, in July, the CP content was 10 g/kg DM greater ($P < 0.001$) in CL swards compared with GR swards. As clover content increased from 18.9% in May to 37.7% in September, the NDF concentration of the CL swards increased by 17.7 g/kg DM, whereas the NDF concentration of the GR swards increased by 49.2 g/kg of DM over the same period. In May and July, the CL swards had a greater proportion ($P < 0.001$) of ADF compared with GR swards, however, in September, the reverse occurred. Ash content in the herbage was significantly greater ($P < 0.001$) for CL compared with GR for all measurement periods.

The chemical composition of the TMR diet offered throughout the experiment is presented in Table 2. In contrast to the pasture treatments, whereby the chemical composition was sensitive to weather, growth stage, and so on, the chemical composition of the TMR diet was similar between measurement periods.

Milk Production and Composition

In May, the TMR-fed cows had significantly higher ($P < 0.001$) daily milk, FPCM, MSol yields and a greater BCS (27.0, 28.8, and 2.13 kg/d, and 3.07, respectively), than both GR and CL cows (Table 3). There was no treatment effect on milk fat or milk lactose content (Table 3). Milk protein content was similar for CL and GR cows, and it was 3.5% ($P < 0.001$) higher than that of TMR cows. The CL cows had greater ($P < 0.001$) daily milk, FPCM, and MSol yields than the GR cows (24.7 vs. 22.9 kg/d, 26.1 vs. 24.3 kg/d, and 1.95 vs. 1.81 kg/d, respectively). The

Table 1. A comparison of sward parameters and chemical composition in grass-only and grass–white clover swards during the May, July, and September DMI estimation periods (mean 2015–2021)

	Diet[1]											
	May measurement period				July measurement period				September measurement period			
Item	GR	CL	SE	*P*-value	GR	CL	SE	*P*-value	GR	CL	SE	*P*-value
Pregrazing herbage mass (kg of DM/ha)	1,455^{a}	1,560^{b}	32.9	<0.05	1,373	1,373	30.2	NS	1,852^{a}	1,915^{b}	30.2	<0.05
Pregrazing sward height (cm)	10.19	9.94	0.22	NS	8	7.64	0.202	NS	9.52	9.49	0.202	NS
Postgrazing sward height (cm)	4.22^{a}	4.08^{b}	0.015	<0.001	4.15	4.13	0.014	NS	4.16^{a}	3.97^{b}	0.014	<0.001
Clover content (%)	—	18.9	0.92	—	—	31.3	0.84	—	—	37.7	0.85	—
OMD[2] (g/kg of DM)	860.90	861.76	1.738	NS	824.73	822.43	1.592	NS	825.28^{a}	821.02^{b}	1.590	<0.001
CP (g/kg of DM)	211.54	207.75	3.489	NS	207.42^{a}	217.32^{b}	3.197	<0.001	240.76	245.16	3.197	NS
NDF (g/kg of DM)	348.99	343.81	2.848	NS	396.10^{a}	363.95^{b}	2.601	<0.001	398.15^{a}	361.51^{b}	2.608	<0.001
ADF (g/kg of DM)	213.65^{a}	219.43^{b}	2.409	<0.001	231.42^{a}	238.40^{b}	2.207	<0.001	253.12^{a}	247.04^{b}	2.211	<0.001
Ash (g/kg of DM)	78.58^{a}	80.44^{b}	0.663	<0.001	85.00^{a}	90.75^{b}	0.608	<0.001	86.53^{a}	94.17^{b}	0.608	<0.001

^{a,b}Means bearing different superscripts within a row differ significantly ($P < 0.05$).

[1]Diets: GR = grass only at 250 kg N/ha; CL = grass-clover at 150 kg N/ha. NS = not significant; — indicates no data recorded, as clover content was only measured for the CL treatment.

[2]OMD = OM digestibility.

remaining 4 parameters (milk fat, milk lactose, BW, and BCS) did not differ between the pasture-fed treatments. The TMR cows were the heaviest and had the highest BCS in May ($P < 0.001$); they were 22.5 and 29.9 kg heavier and had 0.7 and 0.6 higher BCS units than the GR and CL cows, respectively.

In July, the TMR cows had higher ($P < 0.001$) daily milk, FPCM, and MSol yields; BCS; and BW (25.1, 27.3, 2.06 kg/d; 3.10; 552.8 kg, respectively) than the GR and CL cows. Although numerically higher, the milk fat content of the TMR cows was not significantly different to that of the CL cows. There were no differences in milk protein content between treatments (3.59%) in July. The TMR treatment milk had a significantly higher ($P < 0.001$) milk lactose concentration compared with the pasture treatments during the July measurement period (4.80% vs. 4.71%, respectively). Body weight and BCS were greater ($P < 0.001$) in July for the TMR-fed cows compared with the CL and GR, which were not statistically different from one another. The same differences and similarities observed during the May measurement period between CL and GR were also noted in July. Daily milk, FPCM, and MSol yields were significantly higher ($P < 0.001$) for CL in comparison with the GR (19.5 vs. 18.5, 21.3 vs. 19.7, and 1.60 vs. 1.49 kg/d, respectively). There were no differences in milk fat, protein, or lactose content; BW; or BCS between the 2 pasture-fed treatments.

In September, milk, FPCM, and MSol yields; milk fat and lactose content; BW; and BCS were greater ($P < 0.001$) for the cows consuming the TMR diet compared with the cows consuming GR and CL. During this measurement period, the same differences ($P < 0.001$) and similarities between CL and GR as occurred in the July measurement period were again present. Additionally, the CL cows greater BCS (+ 0.04 units) than the GR cows ($P < 0.001$). Year and parity had a significant effect ($P < 0.001$) on milk production during the DMI measurement periods.

DMI and Milk Production Efficiencies

The greatest TDMI was observed for the TMR diet in all 3 measurement periods (20.1, 20.5, and 18.2 kg of DM/cow in May, July, and September, respectively) and was consistently significantly higher than either pasture-based treatment ($P < 0.001$). Cows grazing CL had significantly greater ($P < 0.001$) TDMI compared with the GR cows (Table 4). There was an average increase of +1.03 kg TDMI when cows were consuming the CL pasture compared with the GR. The CL treatment maintained a superior ($P < 0.001$) TDMI compared with the GR treatment across the May, July, and September measurement periods by 0.58, 0.62, and 1.88 kg DM/cow, respectively.

Table 2. Chemical composition of the total TMR diet during the DMI estimation periods (mean 2015–2021)

Item	May	July	September	SE	*P*-value
DM (%)	53.4	56.2	54.7	6.33	NS
CP (g/kg of DM)	136.7	141.6	142.2	13.86	NS
NDF (g/kg of DM)	379.4	376.2	380.46	26.78	NS
Ash (g/kg of DM)	66.2	67.5	64.1	5.81	NS
Starch (%)	26.0	27.2	26.3	1.51	NS
OMD[1] (g/kg of DM)	722.9	731.6	740.0	57.92	NS
Gross energy (MJ/kg of DM)	19.10	19.1	19.1	0.12	NS

[1]OMD = OM digestibility.

Cows consuming the TMR diet had significantly higher ($P < 0.001$) daily energy intake than GR and CL throughout the year whereby their mean energy intake was 17% higher than that of the mean values for the pasture-based cows. The CL and GR fed cows consumed a similar daily energy intake in May and July (19.07 and 17.54 UFL/d, respectively), whereas in September, the CL cows consumed an extra 1.84 UFL/d compared with those offered GR (17.9 vs. 16.1 UFL/d). Daily energy balance was similar for all treatments in May and July, however in September, the CL cows had the greatest energy balance ($P < 0.001$), with a daily energy balance of 1.0 and 1.7 UFL/d more than the GR and TMR cows, respectively. Daily energy balance increased throughout the year in the case of the CL treatment, whereas the GR and TMR cows experienced a decline in energy balance between the July and September DMI estimation periods. There were no treatments in negative energy balance during any of the measurement periods.

Diet had a significant effect on milk production efficiencies ($P < 0.001$) across the 3 different stages of lactation (Table 5). Feeding a TMR diet, resulted in significantly greater ($P < 0.001$) production efficiencies compared with the CL and GR treatments in all measurement periods. Total DMI/100 kg BW, STDMY/100 kg BW, and MSol/100 kg BW for the CL treatment were significantly higher ($P < 0.001$) than the GR treatment (3.35 vs. 3.19 kg, 4.45 vs. 4.15 g, and 327.85 vs. 306.28 g, respectively). No significant differences were observed between the pasture-fed cows for MSol/TDMI (kg). In terms of energetic efficiencies, the energy available for milk production after accounting for maintenance was greatest ($P < 0.001$) for the TMR treatment, intermediate for the CL, and lowest for the GR (14.04 vs. 11.68 vs. 10.73 UFL/d). The TMR-fed cows required less energy ($P < 0.05$) to produce 1 kg STDMY (−0.02 UFL/kg STDMY) compared with the CL treatment, whereas the GR cows required a similar amount of energy to that required by both the TMR and CL cows. The total energy available for MSol production was 25% greater ($P < 0.001$) for the TMR cows compared with the GR and CL cows, which were not significantly differ from each other. Additionally, there was no difference in the energy (UFL) available for MSol production after accounting for maintenance between the 3 treatment groups.

DISCUSSION

Effect of Diet on DMI

Although 2 different methodologies were employed to estimate DMI in this study, the n-alkane technique was applied to the GR and CL groups to accurately determine DMI during grazing, as it accounts for plant–animal interactions. In contrast, the electronic feeding system was used for the TMR group to precisely measure DMI, and the animals consumed a mixture of grass silage, maize silage, and concentrate. This approach was preferred because these feeds may contain low concentrations of n-alkanes, which can introduce measurement errors when using the n-alkane technique (Brosh et al., 2003).

In this study, as part of the experimental design, the cows assigned to the TMR treatment were offered an ad libitum diet fed at a 10% refusal rate over the 6 yr of the study, whereas GR and CL were offered a daily herbage allowance of 17 or 18 kg, depending on concentrate supplementation. Therefore, it is not surprising that the TMR cows had a greater DMI compared with GR and CL. In early lactation, Patton et al. (2008) found that an increased dietary energy intake, which resulted from greater DMI increased the milk yield of TMR-fed cows compared with pasture.

The greater TDMI potential of TMR-fed cows compared with GR and CL cows is likely influenced by one or more of the following factors that affect the cow's ingestive behavior, DM per bite, bite rate, and grazing time (Kolver and Muller, 1998). Kolver and Muller (1998) estimated that if cows spent 510 min grazing per day at a bite rate of 55 bites per minute, cows would have to consume 0.68 g/DM per bite to achieve a daily DMI of 19 kg.

The average total DM content of the TMR diet was 57.2%, whereas the DM content of the CL and GR herbage varied from 15% to 24% (data not shown). Therefore, the TMR cows in the current study had to consume a lower quantity of feed substrate to achieve the desired nutritionally balanced DMI. In contrast, the GR and CL

Table 3. Effect of diet on milk production and cow condition during DMI estimation periods (May, July, and September; mean of 2015–2021)

	Diet[1]														
	May measurement period					July measurement period					September measurement period				
Item[2]	GR	CL	TMR	SE	*P*-value	GR	CL	TMR	SE	*P*-value	GR	CL	TMR	SE	*P*-value
Milk yield (kg/d)	22.9^{a}	24.7^{b}	27.0^{c}	0.38	<0.001	18.5^{a}	19.5^{b}	25.1^{c}	0.35	<0.001	15.8^{a}	16.7^{b}	21.9^{c}	0.35	<0.001
FPCM yield (kg/d)	24.3^{a}	26.1^{b}	28.8^{c}	0.41	<0.001	19.7^{a}	21.3^{b}	27.3^{c}	0.38	<0.001	17.9^{a}	19.2^{b}	25.5^{c}	0.38	<0.001
Milk fat content (%)	4.52	4.47	4.59	0.073	NS	4.5^{a}	4.66^{b}	4.71^{b}	0.068	<0.001	4.97^{a}	4.96^{a}	5.18^{b}	0.070	<0.001
Milk protein content (%)	3.50^{a}	3.51^{a}	3.39^{b}	0.027	<0.001	3.57	3.60	3.59	0.020	NS	3.94^{a}	3.96^{a}	3.84^{b}	0.030	<0.001
Milk lactose content (%)	4.83	4.84	4.87	0.019	NS	4.69^{a}	4.72^{a}	4.80^{b}	0.017	<0.001	4.65^{a}	4.68^{a}	4.74^{b}	0.018	<0.001
Milk solids yield (kg/d)	1.81^{a}	1.95^{b}	2.13^{c}	0.031	<0.001	1.49^{a}	1.60^{b}	2.06^{c}	0.029	<0.001	1.39^{a}	1.48^{b}	1.96^{c}	0.029	<0.001
BCS	3.00^{a}	3.01^{a}	3.07^{b}	0.020	<0.001	3.00^{a}	3.01^{a}	3.10^{b}	0.018	<0.001	2.99^{a}	3.03^{b}	3.15^{c}	0.019	<0.001
BW (kg)	514.3^{a}	506.9^{a}	536.8^{b}	5.79	<0.001	522.3^{a}	522.1^{a}	552.8^{b}	5.65	<0.001	540^{a}	545.3^{a}	570.7^{b}	5.66	<0.001

[a–c]Means bearing different superscripts within a row differ significantly ($P < 0.05$).

[1]Diets: GR = grass only at 250 kg N/ha; CL = grass–white clover at 150 kg N/ha.

[2]FPCM = fat- and protein-corrected milk; BCS = body condition score was assessed on a scale of 1–5, where 1 represented emaciation and 5 represented obesity.

Table 4. Effect of diet on total DMI, energy intake, and energy balance during DMI estimation periods (mean 2015–2021)

	Diet[1]														
	May[2] measurement period					July measurement period					September measurement period				
Item	GR	CL	TMR	SE	*P*-value	GR	CL	TMR	SE	*P*-value	GR	CL	TMR	SE	*P*-value
Total DMI (kg)	16.98^{a}	17.56^{b}	20.06^{c}	0.33	<0.001	16.70^{a}	17.32^{a}	20.49^{b}	0.306	<0.001	15.11^{a}	16.99^{b}	18.21^{c}	0.322	<0.001
Energy intake (UFL[3]/d)	18.74^{a}	19.40^{a}	20.77^{b}	0.341	<0.001	17.23^{a}	17.85^{a}	21.02^{b}	0.316	<0.001	16.08^{a}	17.92^{b}	19.10^{c}	0.332	<0.001
Energy balance (UFL/d)	1.2	0.9	0.9	0.25	NS	1.7	1.4	1.4	0.250	NS	1.1^{a}	2.1^{b}	0.4^{c}	0.2	<0.05

[a–c]Means bearing different superscripts within a row differ significantly ($P < 0.05$).

[1]Diets: GR = grass only at 250 kg N/ha; CL = grass–white clover at 150 kg N/ha.

[2]May = May only represents samples taken in 2015, 2016, 2017, 2019, and 2021.

[3]UFL = unité fourragère lait.

Table 5. Effect of diet on BW, milk, and energetic production efficiencies (mean 2015–2021)

Item[2]	Diet[1]			SE	*P*-value		
	GR	CL	TMR		Diet	Period	Diet × period
BW (kg)	524.6[a]	524.8[a]	553.4[b]	5.38	<0.001	<0.001	<0.05
TDMI (kg/100 kg of BW)	3.19[a]	3.35[b]	3.60[c]	0.042	<0.001	<0.001	<0.05
STDMY (kg/100 kg of BW)	4.15[a]	4.45[b]	5.05[c]	0.062	<0.001	<0.001	<0.001
Milk solids (g/100 kg of BW)	306.28[a]	327.85[b]	371.43[c]	4.640	<0.001	<0.001	<0.001
Milk solids/TDMI (kg)	0.097[a]	0.099[a]	0.107[b]	0.0012	<0.001	<0.001	<0.001
UFL available for milk after maintenance	10.73[a]	11.68[b]	14.04[c]	0.184	<0.001	<0.001	<0.001
UFL used/kg of STDMY	0.52[a]	0.53[a]	0.51[ab]	0.007	<0.05	<0.01	<0.001
Milk solids (g/UFL of NEI)	91.46[a]	92.79[a]	102.75[b]	1.134	<0.001	<0.001	<0.001
Milk solids/UFL available after maintenance (g)	149.52	146.23	150.17	2.220	NS	<0.05	<0.001

[a–c]Means bearing different superscripts within a row differ significantly ($P < 0.05$).

[1]Diets: GR = grass only at 250 kg N/ha; CL = grass–white clover at 150 kg N/ha.

[2]BW = body weight (average) during DMI estimation periods; TDMI = total dry matter intake; STDMY = standard milk (4.0% fat and 3.1% protein); Milk solids = fat (kg) + protein (kg); UFL = unité fourragère lait; NEI = net energy intake.

treatments consumed large quantities of material that included indigestible fiber, which resulted in rumen physical fill, thus reducing the DMI capacity of the cow. In an investigation into the neuroendocrine and physiological factors influencing intake. Roche et al. (2008) reported that in a situation where forage is not limited, cattle consuming grass generally eat for between 4 and 11 h/d compared with TMR-fed cows in a confinement system who eat for a considerably shorter time (4–5 h/d) but feed more often. Vibart et al. (2008) concluded that total nutrient intake was greater for cows in a well-managed confinement system fed a chopped and nutritionally balanced TMR compared with pasture-fed cows who have a more laborious job to manually harvest grass.

The proportion of NDF in the TMR diet may have been a factor influencing TDMI. Neutral detergent fiber only varied by 3 g/kg DM between the DMI measurement periods for the TMR treatment, and the pasture-based treatments had a NDF variation of 54 g/kg DM across measurement periods. Kolver and Muller (1998) reported that grazing cows had a greater NDF intake per kilogram live weight and lower DMI compared with TMR-fed cows (1.5% vs. 4.2% and 19.0% vs. 23.4 kg, respectively). The differences in DMI and subsequently milk production are most likely affected by energy maintenance requirements and differences in energy intake (Bargo et al., 2002).

The greater milk production associated with the inclusion of white clover in the sward is generally associated with higher voluntary DMI, as observed in this experiment, and improved nutritive value of the herbage compared with grass-only swards (Clark and Harris, 1996; Harris et al., 1998; Egan et al., 2018). In the current study, TDMI was 1.03 kg DM/cow greater in CL compared with GR swards at an average sward clover content of 29%. Ribeiro Filho et al. (2003), Egan et al. (2018), and McClearn et al. (2021) found that when sward white clover content was 40%, 23%, and 19%, respectively, DMI of cows grazing clover swards increased by 1.50, 0.90, and 0.80 kg DM/cow per day, respectively, compared with cows consuming PRG-only swards. In contrast, Enriquez-Hidalgo et al. (2014a) found no increase in DMI between cows consuming white clover and PRG-only swards at a sward white clover content of 20%. The CL cows in this study, similar to McClearn et al. (2019b) and Murray et al. (2024), grazed lower into the sward compared with the GR during the DMI estimation periods; this combined with the lower NDF values of the CL herbage indicates greater palatability, which subsequently gave rise to greater DMI. This was particularly obvious in September when the sward white clover content was at its greatest (38%) and the NDF content of the CL sward was 36 g/kg DM lower than the GR sward.

Grazing cattle have a 70% partial preference for white clover compared with PRG only when both sward types are available (Rutter, 2006). This is generally as a result of a faster rate of rumen passage associated with white clover because of its lower fiber content making it easier to break down in the rumen compared with PRG (Beever et al., 1985; Dewhurst et al., 2003). Moseley and Jones (1984) reported that the percentage of rumen particulate matter disappearing in the rumen during the first 3 h after feeding was 57% for white clover compared with 49% for PRG. This enables the cow grazing white clover swards to consume a greater daily quantity of herbage, as in the present study. According to Enriquez-Hidalgo et al. (2014b), rumination time was reduced when sward white clover content was high (31%) compared with PRG-only swards (395 vs. 463 min/d), indicating a greater ease of digestion and increasing voluntary DMI.

Similar to the comparison between the DMI differences between the TMR and pasture treatments, the most common physical factors influencing DMI are dietary fiber content, the digestibility of that fiber, and the degradation rate of that fiber in the rumen (Roche et al., 2008). It has been suggested that the NDF fiber fraction is most likely to influence DMI compared with the other herbage chemical fractions (Van Soest, 1994). Allen (2000) reported experimental variations in the relationship between rumen physical fill and DMI. In an experiment conducted by Dado and Allen (1995), a decline in DMI was observed when cows were fed a diet containing 35% NDF, whereas when the diet contained 25% NDF no decline in DMI was observed. In contrast, Beecher et al. (2018) reported no relationship between voluntary DMI and herbage NDF content. In the current study, the NDF content of the white clover swards was on average 25 g/kg lower than that of the GR swards on average over the 3 measurement periods. This, combined with the 1.03 kg/cow increase in TDMI indicates that the lower NDF content encouraged a greater DMI intake for cows consuming white clover swards. Egan et al. (2018) and McClearn et al. (2021) also reported additional DMI for cows consuming white clover when the NDF content was lower compared with cows consuming PRG-only swards. Further, more complex investigations into the relationship between NDF digestion and DMI are required to draw a definite conclusion on the mechanisms associated with greater DMI when cows consume white clover swards. Investigating the undigested amylase- and sodium sulphite-treated NDF corrected for ash residue to predict digestibility and the nutritive value of herbage as described by Dineen et al. (2020, 2021) may be beneficial to better understand the relationship between fodder digestibility and DMI.

Milk Production

Similar to previous studies (e.g., Kolver and Muller, 1998; Bargo et al., 2002), daily milk and MSol yields were greater for TMR cows than the CL and GR cows. It has long been reported that DMI is the reason for the lower milk production of high-producing dairy cows when fed high-quality grass compared with those fed a TMR diet (Kolver and Muller, 1998; O'Neill et al., 2011). Available dietary energy is also a primary limiting factor for high-producing cows fed pasture compared with those fed TMR (Bargo et al., 2002). The TMR diet in this experiment was generated to support 2.5 kg of MSol/d at peak milk production and the components were balanced for CP, fiber, and energy. In contrast, the GR and CL diets, and consequently, milk production, were variable due to fluctuations in grass quality and supply throughout the grazing season (Hennessy et al., 2020). Therefore, offering a TMR diet that is fully nutritionally balanced, similar to the one in the current study, will support a high DMI, which in turn results in greater milk, FPCM, and MSol yields compared with a pasture-based diet.

Daily milk, FPCM, and MSol yields were greater for CL compared with GR during the DMI measurement periods, similar to Egan et al. (2018) and McClearn et al. (2021). Egan et al. (2018) achieved a high average annual sward white clover content (24.6%) and found that differences in milk production between grass-only and grass–white clover swards were observed from June onward when the DMI of cows grazing swards containing white clover was greater than that of cows grazing PRG-only swards. However, the current study found an increase in daily milk and MSol yields in May at a sward white clover inclusion rate of 18.9%, and McClearn et al. (2021) observed increased milk production in the summer measurement period at a 14.2% sward white clover content. Additionally, Murray et al. (2024) achieved a significant increase in milk response when white was included in the sward, at a white clover inclusion rate of 18.1%. This suggests that increased milk production can be achieved from well-managed grass-clover swards at lower levels of white clover inclusion than indicated in reviews by Andrews et al. (2007) and Dineen et al. (2018). Sward clover content in combination with herbage quality, and in particular fiber (NDF content), are likely important drivers of increased DMI and subsequently milk production on grass–white clover swards compared with grass-only swards, rather than just the sward clover content.

In the present study, the milk protein concentration was greater for the pasture-fed cows compared with the cows consuming the TMR diet. This can be rationalized by the presence of rapidly fermentable dietary carbohydrates (Jenkins and McGuire, 2006) in the herbage compared with the TMR diet. Consequently, the higher milk protein concentration for the GR and CL, observed in the current study, may have occurred because of the greater digestibility of the pasture-based diets compared with the TMR diet (OMD 836 vs. 709 g/kg DM, respectively). Similar results were also achieved in other studies carried out at the same experimental site (Gulati et al., 2018; O'Callaghan et al., 2016). No difference in milk fat concentration was observed between the TMR and pasture-fed cows in May. This agrees with Allen (1997), who stated that milk fat is more responsive to diet later in lactation compared with early lactation. O'Neill et al. (2011) found no significant differences in milk fat concentrations in a comparison between cows consuming TMR and grazed grass in early lactation. For both the pasture and TMR-fed cows in the current study, fiber was not a factor limiting milk production and all diets had a NDF content exceeding the requirement of a lactating dairy cow of 300 g/kg of DM (Moran, 2005). McAuliffe et al. (2022) found that TMR-fed cows had the greatest proportion of rumen acetic acid, compared with pasture-fed cows. This, combined with the fact that high forage diets, such as the one in the current study, encourage higher rates of saliva production and better rumen buffering that can support higher milk fat concentration, as seen for the TMR cows on average over the 3 DMI measurement periods compared with GR and CL.

The similar milk fat contents observed in May and September in the current experiment, agrees with previous studies (Enriquez-Hidalgo et al., 2014a; Egan et al., 2018). It has been reported that with a white clover content of between 22% and 42%, similar milk fat contents can be achieved from cows grazing PRG–white clover and PRG-only (Leach et al., 2000; Phillips et al., 2000; Ribeiro Filho et al., 2005). However, the presence of white clover in the sward does have the potential to decrease milk fat content (Thomson et al., 1985; Harris et al., 1997), as reported during July in the current study. This is generally attributed to a faster rate of rumen passage of the grass–white clover swards compared with PRG alone (Thomson et al., 1985). In July, the CP content in the herbage was significantly ($P < 0.001$) greater than the GR. During the same measurement period, there was no difference detected between the protein content of the milk. This substantiates the fact that when CP levels are in excess for dairy cow maintenance plus milk production capacity requirements (150–180 g/kg of DM; NRC, 2001), feeding more CP can't provide any further milk production benefit. Both treatments contained herbage CP concentrations greater than the requirements of grazing ruminants, which can lead to an inefficient utilization of dietary N and N losses to the environment (van Dorland et al., 2007).

Energy Balance and Production Efficiencies

The TMR-fed cows exhibited superior milk production efficiencies, consuming a greater quantity of TDMI per 100 kg of BW and producing more MSol per kilogram TDMI compared with the grass-based treatments. This can further be substantiated by the TMR diet's ability to support a greater BW, and produce greater quantities of STDMY and MS per 100 kg of BW above the CL and GR diets.

In the present study, the TMR cows had a 3.33 and 2.30 kg cow/d, respectively, greater DMI, and a 31% and 22%, respectively, greater MSol production compared with the GR and CL treatments. This derived from the significantly greater proportion of dietary energy available for intake and greater energy available after accounting for maintenance for milk production, compared with the pasture cows. Additionally, the TMR cows had 12% greater total energy available for MS production compared with the pasture-based cows, and required a 4% lower UFL energy requirement to produce 1 kg of STDMY to that required by the CL cows. These findings agree with previous studies, which also found an increase in DMI and milk production when cows were fed TMR compared with pasture (Kolver and Muller, 1998; Bargo et al., 2002) due to greater available dietary energy.

In terms of milk production efficiencies for the pasture-based cows, the inclusion of white clover in the sward resulted in greater STDMY per 100 kg of BW and MSol per 100 kg of BW compared with PRG only swards, despite similar BW. This can most likely be explained by the higher milk production and greater sward nutritive quality, similar to the results reported by McClearn et al. (2021). Despite both treatments having a similar BW, the CL cows had 9% more energy available for milk production after maintenance compared with the GR cows. This greater dilution of maintenance energy requirements by the CL cows resulted in greater MSol per 100 kg of BW and MSol per TDMI over the PRG-only diet. Nonetheless, although greater energy was available for milk production for the CL treatment, both pasture-based treatments had similar UFL requirements to produce 1 kg of STDMY and 1 kg of MSol. Thus, the additional available energy from the CL diet did not improve the feed conversion efficiency of the diet in comparison with GR. Coffey et al. (2017) and Prendiville et al. (2009) observed that the energy requirements for milk production can be reduced when cross bred (Jersey × Holstein-Friesian) animals were used in an intensive grazing system. Therefore, further investigation into the relationship between breed and sward type may be warranted to improve feed and milk production efficiencies in pasture-based dairy production. These are important metrics because in intensive grass-based dairy production systems, cows must be capable of consuming high quantities of good quality pasture per unit of BW and efficiently convert this feed into high-quality MSol (Buckley et al., 2005). Daily energy intake was 6% greater for the CL diet on average over the measurement periods, at a similar daily pasture allowance to the GR treatment. This indicates that white clover inclusion in the sward has a greater potential to meet the energy requirements for pasture-fed dairy cows to maximize the animals' genetic potential for milk yield.

CONCLUSIONS

The present study shows that animal performance and production efficiencies were influenced by dairy cow diet. Feeding a TMR diet significantly increased DMI, milk, and MSol yield compared with the pasture-based diets, which was a reflection of the greater energy available for milk production after maintenance. Including white clover in the diet of dairy cows, when it is present in the sward at sufficient levels, can increase MSol production through increased DMI, despite similar amounts of energy being available for milk production on both pasture-based treatments. This experiment demonstrated that different feeding strategies such as a TMR feeding regimen or incorporating white clover into the sward can effectively be used as mechanisms to raise milk production, TDMI, and production efficiencies above a PRG-only diet. Some of the main factors influencing DMI and subsequently, production efficiency, include nutritive quality of the diet, greater voluntary intake and digestibility of the fiber consumed. Thus, further and more complex investigations into feed quality and N use efficiency, could provide the opportunity to improve the efficiency, profitability, and sustainability of ruminant production systems.

NOTES

This research was funded by the Irish Dairy Levy administered by Dairy Research Ireland. The first author received a Teagasc Walsh Scholarship, Teagasc, Oak Park, Carlow, Ireland. The authors gratefully acknowledge the assistance and care of the animals provided by the technical and farm staff at Teagasc, Animal and Grassland Research and Innovation Centre, Moorepark, Fermoy, Co. Cork. All experimental procedures on the animals in this study were approved by the Health Products and Regulatory Authority of Ireland under the project authorizations AE19132/P019, AE19132/P095, and AE19132/P138. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: CL = grass-clover swards receiving 150 kg N/ha; FPCM = fat- and protein-corrected milk; GR = grass-only swards receiving 250 kg N/ha; MSol = milk solids; NEI = net energy intake; NS = not significant; OMD = OM digestibility; PBI = PastureBase Ireland; PRG = perennial ryegrass; STDMY = standard milk yield; TDMI = total DMI; UFL = unité fourragère lait.

REFERENCES

Allen, M. S. 1997. Relationship between fermentation acid production in the rumen and the requirement for physically effective fiber. J. Dairy Sci. 80:1447–1462. https://doi.org/10.3168/jds.S0022-0302(97)76074-0.

Allen, M. S. 2000. Effects of diet on short-term regulation of feed intake by lactating dairy cattle. J. Dairy Sci. 83:1598–1624. https://doi.org/10.3168/jds.S0022-0302(00)75030-2.

Andrews, M., D. Scholefield, M. Abberton, B. McKenzie, S. Hodge, and J. Raven. 2007. Use of white clover as an alternative to nitrogen fertiliser for dairy pastures in nitrate vulnerable zones in the UK: productivity, environmental impact and economic considerations. Ann. Appl. Biol. 151:11–23. https://doi.org/10.1111/j.1744-7348.2007.00137.x.

AOAC. 1990. Official Methods of Analysis. Vol. 1. 15th ed. AOAC, Washington, DC.

Bargo, F., L. D. Muller, J. E. Delahoy, and T. W. Cassidy. 2002. Performance of high producing dairy cows with three different feeding systems combining pasture and total mixed rations. J. Dairy Sci. 85:2948–2963. https://doi.org/10.3168/jds.s0022-0302(02)74381-6.

Bargo, F., L. D. Muller, E. S. Kolver, and J. E. Delahoy. 2003. Invited review: Production and digestion of supplemented dairy cows on pasture. J. Dairy Sci. 86:1–42. https://doi.org/10.3168/jds.S0022-0302(03)73581-4.

Beecher, M., R. Baumont, M. O'Donovan, T. M. Boland, J. Aufrere, C. Fleming, N. Galvin, and E. Lewis. 2018. Effects of harvesting perennial ryegrass at different levels of herbage mass on voluntary intake and in vivo digestibility in sheep. Grass Forage Sci. 73:553–561. https://doi.org/10.1111/gfs.12319.

Beever, D. E., D. J. Thomson, M. J. Ulyatt, S. B. Cammell, and M. C. Spooner. 1985. The digestion of fresh perennial ryegrass (*Lolium perenne* L. cv. Melle) and white clover (*Trifolium repens* L. cv. Blanca) by growing cattle fed indoors. Br. J. Nutr. 54:763–775. https://doi.org/10.1079/bjn19850162.

Brosh, A., Z. Henkin, S. J. Rothman, Y. Aharoni, A. Orlov, and A. Arieli. 2003. Effects of faecal n-alkane recovery in estimates of diet composition. J. Agric. Sci. 140:93–100. https://doi.org/10.1017/S0021859602002757.

Buckley, F., C. Holmes, and M. G. Keane. 2005. Genetic characteristics required in dairy and beef cattle for temperate grazing systems. Pages 61–79 in Utilization of Grazed Grass in Temperate Animal Systems. Proc. Satellite Workshop XXth Int. Grassl. Congr., Cork, Ireland.

Charlton, G. L., S. M. Rutter, M. East, and L. A. Sinclair. 2011. Preference of dairy cows: Indoor cubicle housing with access to a total mixed ration vs. access to pasture. Appl. Anim. Behav. Sci. 130:1–9. https://doi.org/10.1016/j.applanim.2010.11.018.

Clark, D. A., and S. L. Harris. 1996. White clover or nitrogen fertiliser for dairying? Agronomy Society of New Zealand Special Publication No. 11. Grassl. Res. ad Pract. Ser. 6:107–114.

Coffey, E. L., L. Delaby, S. Fitzgerald, N. Galvin, K. M. Pierce, and B. Horan. 2017. Effect of stocking rate and animal genotype on dry matter intake, milk production, body weight, and body condition score in spring-calving, grass-fed dairy cows. J. Dairy Sci. 100:7556–7568. https://doi.org/10.3168/jds.2017-12672.

Dado, R. G., and M. S. Allen. 1995. Intake limitations, feeding behavior, and rumen function of cows challenged with rumen fill from dietary fiber or inert bulk. J. Dairy Sci. 78:118–133. https://doi.org/10.3168/jds.S0022-0302(95)76622-X.

Dewhurst, R. J., R. T. Evans, N. D. Scollan, J. M. Moorby, R. J. Merry, and R. J. Wilkins. 2003. Comparison of grass and legume silages for milk production. 2. In vivo and in sacco evaluations of rumen function. J. Dairy Sci. 86:2612–2621. https://doi.org/10.3168/jds.S0022-0302(03)73856-9.

Dillon, P., S. Crosse, G. Stakelum, and F. Flynn. 1995. The effect of calving date and stocking rate on the performance of spring-calving dairy cows. Grass Forage Sci. 50:286–299. https://doi.org/10.1111/j.1365-2494.1995.tb02324.x.

Dillon, P. G. 1993. The use of n-alkanes as markers to determine herbage intake, botanical composition of available or consumed herbage and in studies of digesta kinetics with dairy cows. PhD diss. National University of Ireland.

Dineen, M., L. Delaby, T. Gilliland, and B. McCarthy. 2018. Meta-analysis of the effect of white clover inclusion in perennial ryegrass swards on milk production. J. Dairy Sci. 101:1804–1816. https://doi.org/10.3168/jds.2017-12586.

Dineen, M., B. McCarthy, P. Dillon, P. A. LaPierre, S. Fessenden, C. Matthews, N. Galvin, and M. E. Van Amburgh. 2020. Rumen metabolism, omasal flow of nutrients, and microbial dynamics in lactating dairy cows fed fresh perennial ryegrass (*Lolium perenne* L.) not supplemented or supplemented with rolled barley grain. J. Dairy Sci. 103:11332–11348. https://doi.org/10.3168/jds.2020-18437.

Dineen, M., B. McCarthy, D. Ross, A. Ortega, P. Dillon, and M. E. Van Amburgh. 2021. Characterization of the nutritive value of perennial ryegrass (*Lolium perenne* L.) dominated pastures using updated chemical methods with application for the Cornell Net Carbohydrate and Protein System. Anim. Feed Sci. Technol. 272:114752. https://doi.org/10.1016/j.anifeedsci.2020.114752.

Edmundson, A. J., I. J. Lean, L. D. Weaver, T. Farver, and G. Webster. 1989. A body condition scoring chart for Holstein dairy cows. J. Dairy Sci. 72:68–78. https://doi.org/10.3168/jds.S0022-0302(89)79081-0.

Egan, M., N. Galvin, and D. Hennessy. 2018. Incorporating white clover (*Trifolium repens* L.) into perennial ryegrass (*Lolium perenne* L.) swards receiving varying levels of nitrogen fertilizer: Effects on milk and herbage production. J. Dairy Sci. 101:3412–3427. https://doi.org/10.3168/jds.2017-13233.

Enriquez-Hidalgo, D., T. Gilliland, M. H. Deighton, M. O'Donovan, and D. Hennessy. 2014b. Milk production and enteric methane emissions by dairy cows grazing fertilized perennial ryegrass pasture with or without inclusion of white clover. J. Dairy Sci. 97:1400–1412. https://doi.org/10.3168/jds.2013-7034.

Enriquez-Hidalgo, D., D. Hennessy, T. Gilliland, M. Egan, J. F. Mee, and E. Lewis. 2014a. Effect of rotationally grazing perennial ryegrass white clover or perennial ryegrass only swards on dairy cow feeding behaviour, rumen characteristics and sward depletion patterns. Livest. Sci. 169:48–62. https://doi.org/10.1016/j.livsci.2014.09.002.

European Commission. 1991. Council Directive 91/676/EEC of 12 December 1991 concerning the protection of waters against pollution caused by nitrates from agricultural sources. Official Journal 1991:L 375 [31/12/1991]. Accessed May 18, 2026. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A01991L0676-20260302.

Evers, S. H., L. Delaby, K. M. Pierce, B. McCarthy, E. L. Coffey, and B. Horan. 2023. An evaluation of detailed animal characteristics influencing the lactation production efficiency of spring-calving, pasture-based dairy cattle. J. Dairy Sci. 106:1097–1109. https://doi.org/10.3168/jds.2022-21815.

Faverdin, P., C. Baratte, R. Delagarde, and J. L. Peyraud. 2011. GrazeIn: a model of herbage intake and milk production for grazing dairy cows. 1. Prediction of intake capacity, voluntary intake and milk production during lactation. Grass Forage Sci. 66:29–44. https://doi.org/10.1111/j.1365-2494.2010.00776.x.

Faverdin, P., R. Delagarde, L. Delaby, and F. Meschy. 2007. Alimentation des vaches laitières. Pages 23–55 in Alimentation des bovins, ovins et caprins. QUAE Editions, INRA, Versailles. France.

Ganche, E., L. Delaby, M. O'Donovan, T. M. Boland, and E. Kennedy. 2013. Direct and carryover effect of post-grazing sward height on total lactation dairy cow performance. Animal 7:1390–1400. https://doi.org/10.1017/S1751731113000451.

Gulati, A., N. Galvin, E. Lewis, D. Hennessy, M. O'Donovan, J. J. McManus, M. A. Fenelon, and T. P. Guinee. 2018. Outdoor grazing of dairy cows on pasture versus indoor feeding on total mixed ration: Effects on gross composition and mineral content of milk during lactation. J. Dairy Sci. 101:2710–2723. https://doi.org/10.3168/jds.2017-13338.

Hanrahan, L., A. Geoghegan, M. O'Donovan, V. Griffith, E. Ruelle, M. Wallace, and L. Shalloo. 2017. PastureBase Ireland: A grassland decision support system and national database. Comput. Electron. Agric. 136:193–201. https://doi.org/10.1016/j.compag.2017.01.029.

Harris, S. L., M. J. Auldist, D. A. Clark, and E. B. Jansen. 1998. Effects of white clover content in the diet on herbage intake, milk production and milk composition of New Zealand dairy cows housed indoors. J. Dairy Res. 65:389–400. https://doi.org/10.1017/s0022029998002969.

Harris, S., D. Clark, and E. Jansen. 1997. Optimum white clover content for milk production. Pages 169-171 in Proc. NZ Society of Animal Production. New Zealand Society of Animal Production. https://www.nzsap.org/system/files/proceedings/1997/ab97047.pdf.

Hennessy, D., L. Delaby, A. van den Pol-van Dasselaar, and L. Shalloo. 2020. Increasing grazing in dairy cow milk production systems in Europe. Sustainability (Basel) 12:2443. https://doi.org/10.3390/su12062443.

INRA. 2010. Alimentation des bovins, ovins et caprins: Besoins des animauxValeurs des aliments Tables Inra 2010 édition remaniée. Page 315. Editions Quae.

Jenkins, T. C., and M. A. McGuire. 2006. Major advances in nutrition: Impact on milk composition. J. Dairy Sci. 89:1302–1310. https://doi.org/10.3168/jds.S0022-0302(06)72198-1.

Kolver, E. S., and L. D. Muller. 1998. Performance and nutrient intake of high producing Holstein cows consuming pasture or a total mixed ration. J. Dairy Sci. 81:1403–1411. https://doi.org/10.3168/jds.S0022-0302(98)75704-2.

Krause, K. M., and G. R. Oetzel. 2006. Understanding and preventing subacute ruminal acidosis in dairy herds: A review. Anim. Feed Sci. Technol. 126:215–236. https://doi.org/10.1016/j.anifeedsci.2005.08.004.

Leach, K. A., J. A. Bax, D. J. Roberts, and C. Thomas. 2000. The establishment and performance of a dairy system based on perennial ryegrass—White clover swards compared with a system based on nitrogen fertilized grass. Biol. Agric. Hortic. 17:207–227. https://doi.org/10.1080/01448765.2000.9754843.

Lorenz, H., T. Reinsch, S. Hess, and F. Taube. 2019. Is low-input dairy farming more climate friendly? A meta-analysis of the carbon footprints of different production systems. J. Clean. Prod. 211:161–170. https://doi.org/10.1016/j.jclepro.2018.11.113.

Macdonald, K. A., D. Beca, J. W. Penno, J. A. S. Lancaster, and J. R. Roche. 2011. Effect of stocking rate on the economics of pasture-based dairy farms. J. Dairy Sci. 94:2581–2586. https://doi.org/10.3168/jds.2010-3688.

Macdonald, K. A., J. W. Penno, J. A. S. Lancaster, A. M. Bryant, J. M. Kidd, and J. R. Roche. 2017. Production and economic responses to intensification of pasture-based dairy production systems. J. Dairy Sci. 100:6602–6619. https://doi.org/10.3168/jds.2016-12497.

Mayes, R. W., C. S. Lamb, and P. M. Colgrove. 1986. The use of dosed and herbage n-alkanes as markers for the determination of herbage intake. J. Agric. Sci. 107:161–170. https://doi.org/10.1017/s0021859600066910.

McAuliffe, S., J. F. Mee, E. Lewis, N. Galvin, and D. Hennessy. 2022. Feeding system effects on dairy cow rumen function and milk production. Animals (Basel) 12:523. https://doi.org/10.3390/ani12040523.

McClearn, B., L. Delaby, T. J. Gilliland, C. Guy, M. Dineen, F. Coughlan, N. Galvin, and B. McCarthy. 2021. The effect of *Lolium perenne* L. ploidy and *Trifolium repens* L. inclusion on dry matter intake and production efficiencies of spring-calving grazing dairy cows. J. Dairy Sci. 104:6688–6700. https://doi.org/10.3168/jds.2020-19753.

McClearn, B., T. Gilliland, C. Guy, M. Dineen, F. Coughlan, and B. McCarthy. 2019a. The effect of perennial ryegrass ploidy and white clover inclusion on milk production of dairy cows. Anim. Prod. Sci. 60:143–147. https://doi.org/10.1071/AN18539.

McClearn, B., T. J. Gilliland, L. Delaby, C. Guy, M. Dineen, F. Coughlan, and B. McCarthy. 2019b. Milk production per cow and per hectare of spring-calving dairy cows grazing swards differing in *Lolium perenne* L. ploidy and *Trifolium repens* L. composition. J. Dairy Sci. 102:8571–8585. https://doi.org/10.3168/jds.2018-16184.

Minson, D. J. 1990. Forage in Ruminant Nutrition. Academic Press Ltd., London, UK. https://scholar.google.com/scholar_lookup?title=Forage%20in%20Ruminant%20Nutrition&publication_year=1990&author=D.J.%20Minson.

Moran, J. 2005. Tropical dairy farming: Feeding management for small holder dairy farmers in the humid tropics. Landkinks Press, Clayton, Australia. https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q= Moran%2C+J.+2005.+Tropical+dairy+farming%3A+feeding+management+for+small+holder+dairy+farmers+in+the+humid+tropics. +Landkinks+Press%2C+Clayton%2C+Australia.&btnG=.

Morgan, D. J., G. Stakelum, and J. Dwyer. 1989. Modified neutral detergent cellulase digestibility procedure for use with the 'Fibertec' system. Irish J. Agric. Research. 28:91–92. https://www.cabidigitallibrary.org/doi/full/10.5555/19891420119.

Moseley, G., and J. R. Jones. 1984. The physical digestion of perennial ryegrass (*Lolium perenne*) and white clover (*Trifolium repens*) in the foregut of sheep. Br. J. Nutr. 52:381–390. https://doi.org/10.1079/BJN19840104.

Murray, Á., L. Delaby, T. J. Gilliland, and B. McCarthy. 2024. Nitrogen offset potential in a multiyear farmlet-scale study: Milk and herbage production from grazed perennial ryegrass–white clover swards. J. Dairy Sci. 107:2129–2142. https://doi.org/10.3168/jds.2023-23822.

NRC. 2001. Nutrient Requirements of Dairy Cattle. 7th rev. ed. National Academies Press, Washington. DC. https://www.nationalacademies.org/read/9825.

O'Callaghan, T. F., D. Hennessy, S. McAuliffe, K. N. Kilcawley, M. O'Donovan, P. Dillon, R. P. Ross, and C. Stanton. 2016. Effect of pasture versus indoor feeding systems on raw milk composition and quality over an entire lactation. J. Dairy Sci. 99:9424–9440. https://doi.org/10.3168/jds.2016-10985.

O'Donovan, M., J. Connolly, P. Dillon, M. Rath, and G. Stakelum. 2002. Visual assessment of herbage mass. Ir. J. Agric. Food Res. 41:201–211. https://www.jstor.org/stable/25562464.

O'Donovan, M., E. Lewis, and P. O'Kiely. 2011. Requirements of future grass-based ruminant production systems in Ireland. Ir. J. Agric. Food Res. 50:1–21. https://www.jstor.org/stable/41348152.

O'Neill, B. F., M. H. Deighton, B. M. O'Loughlin, F. J. Mulligan, T. M. Boland, M. O'Donovan, and E. Lewis. 2011. Effects of a perennial ryegrass diet or total mixed ration diet offered to spring-calving Holstein-Friesian dairy cows on methane emissions, dry matter intake, and milk production. J. Dairy Sci. 94:1941–1951. https://doi.org/10.3168/jds.2010-3361.

Patton, J., G. Olmos, and J. J. Murphy. 2008. Comparison of total mixed ration and pasture feeding systems for early lactation dairy cows. Page 30 in Proc. Agric. Res. Forum. https://www.researchgate.net/profile/Gabriela-Olmos-Antillon/publication/293125435_Comparison_of_total_mixed_ration_and_pasture_feeding_systems_for_early_lactation_dairy_cows/links/56b9c8f108ae7e3a0fa0886c/Comparison-of-total-mixed-ration-and-pasture-feeding-systems-for-early-lactation-dairy-cows.pdf.

Peyraud, J. L., A. Van den Pol, P. Dillon, and L. Delaby. 2010. Producing milk from grazing to reconcile economic and environmental performances. Pages 163–164 in In 23th General Meeting of the European Grassland Federation, Kiel, Germany. Accessed Feb. 19, 2022. https://www.europeangrassland.org/fileadmin/documents/Infos/Printed_Matter/Proceedings/EGF2010_GSE_vol15.pdf#page=205.

Phillips, C. J. C., N. L. James, and H. M. Nyallu. 2000. The effects of forage supplements on the ingestive behaviour and production of dairy cows grazing ryegrass only or mixed ryegrass and white clover pastures. Anim. Sci. 70:555–559. https://doi.org/10.1017/S1357729800051900.

Prendiville, R., K. M. Pierce, and F. Buckley. 2009. An evaluation of production efficiencies among lactating Holstein-Friesian, Jersey, and Jersey × Holstein-Friesian cows at pasture. J. Dairy Sci. 92:6176–6185. https://doi.org/10.3168/jds.2009-2292.

Ribeiro Filho, H. M. N., R. Delagarde, and J. L. Peyraud. 2003. Inclusion of white clover in strip-grazed perennial ryegrass swards: Herbage intake and milk yield of dairy cows at different ages of sward regrowth. Anim. Sci. 77:499–510. https://doi.org/10.1017/S1357729800054448.

Ribeiro Filho, H. M. N., R. Delagarde, and J. L. Peyraud. 2005. Herbage intake and milk yield of dairy cows grazing perennial ryegrass swards or white clover/perennial ryegrass swards at low-and medium-herbage allowances. Anim. Feed Sci. Technol. 119:13–27. https://doi.org/10.1016/j.anifeedsci.2004.12.009.

Roche, J. R., D. Blache, J. K. Kay, D. R. Miller, A. J. Sheahan, and D. W. Miller. 2008. Neuroendocrine and physiological regulation of intake with particular reference to domesticated ruminant animals. Nutr. Res. Rev. 21:207–234. https://doi.org/10.1017/S0954422408138744.

Rutter, S. M. 2006. Diet preference for grass and legumes in free-ranging domestic sheep and cattle: Current theory and future application. Appl. Anim. Behav. Sci. 97:17–35. https://doi.org/10.1016/j.applanim.2005.11.016.

Sjaunja, L. O. 1990. A Nordic proposal for an energy-corrected milk (ECM) formula. Pages 2–6 in 27th Session International Committee for Recording and Productivity of Milk Animals, Paris, France.

Thomson, D. J., D. E. Beever, M. J. Haines, S. B. Cammell, R. T. Evans, M. S. Dhanoa, and A. R. Austin. 1985. Yield and composition of milk from Friesian cows grazing either perennial ryegrass or white clover in early lactation. J. Dairy Res. 52:17–31. https://doi.org/10.1017/S0022029900023852.

van Dorland, H. A., H. R. Wettstein, H. Leuenberger, and M. Kreuzer. 2007. Effect of supplementation of fresh and ensiled clovers to ryegrass on nitrogen loss and methane emission of dairy cows. Livest. Sci. 111:57–69. https://doi.org/10.1016/j.livsci.2006.11.015.

Van Soest, P. J. 1994. Nutritional Ecology of the Ruminant. Cornell University Press.

Van Soest, P. J., J. B. Robertson, and B. A. Lewis. 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. J. Dairy Sci. 74:3583–3597. https://doi.org/10.3168/jds.S0022-0302(91)78551-2.

Vibart, R. E., V. Fellner, J. C. Burns, G. B. Huntington, and J. T. Green Jr. 2008. Performance of lactating dairy cows fed varying levels of total mixed ration and pasture. J. Dairy Res. 75:471–480. https://doi.org/10.1017/S0022029908003361.

Zebeli, Q., J. Dijkstra, M. Tafaj, H. Steingass, B. N. Ametaj, and W. Drochner. 2008. Modeling the adequacy of dietary fiber in dairy cows based on the response of ruminal pH and milk fat production to composition of the diet. J. Dairy Sci. 91:2046–2066. https://doi.org/10.3168/jds.2007-0572.

ORCIDS

E. Fitzpatrick, https://orcid.org/0009-0007-7866-1510
L. Delaby, https://orcid.org/0000-0002-9805-4108
T. J. Gilliland, https://orcid.org/0000-0002-4514-1435
D. Hennessy https://orcid.org/0000-0002-7375-3754